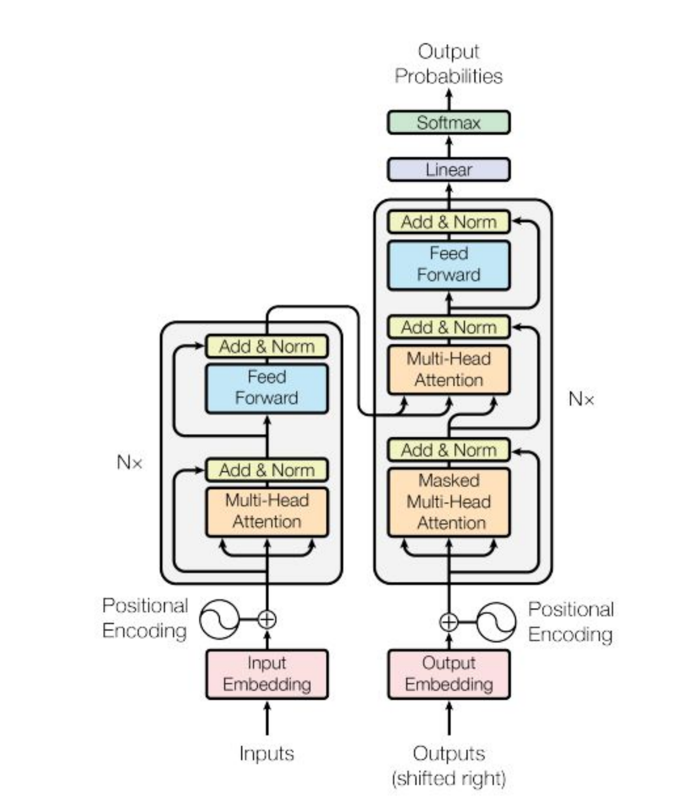
GPT e Attention

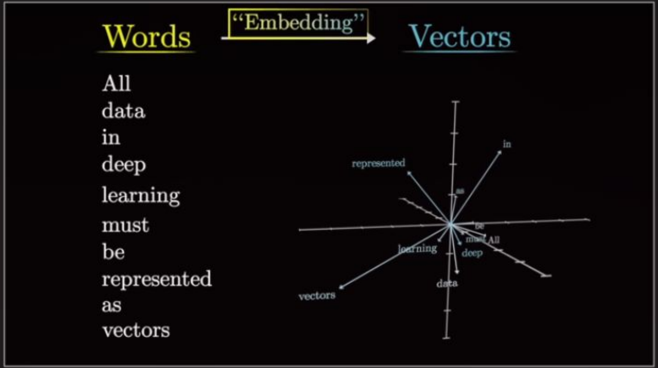
- Predição de próxima palavra
- Token
- Embedding
- Attention
- Exemplos de word embedding
- GPT2



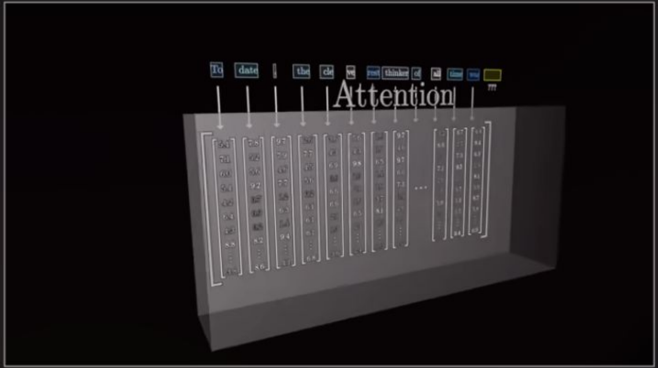
Referencias

- https://arxiv.org/pdf/1706.03762
- https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi
- https://www.youtube.com/watch?v=kCc8FmEb1nY
- https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ

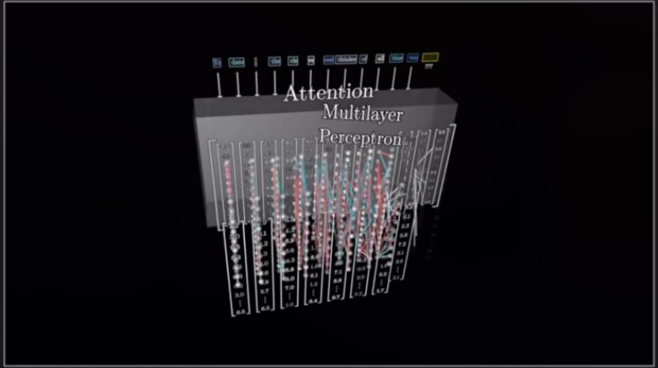
Embedding



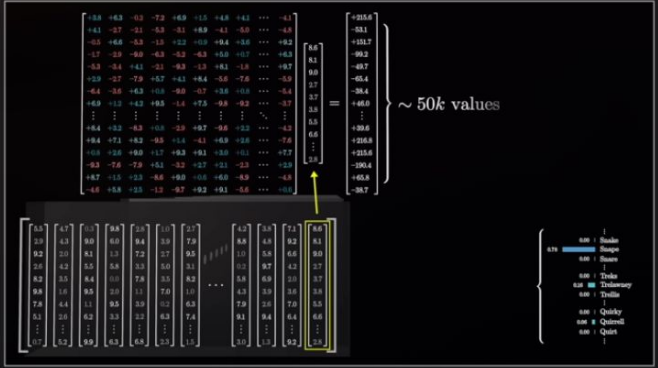
Attention



MLPs



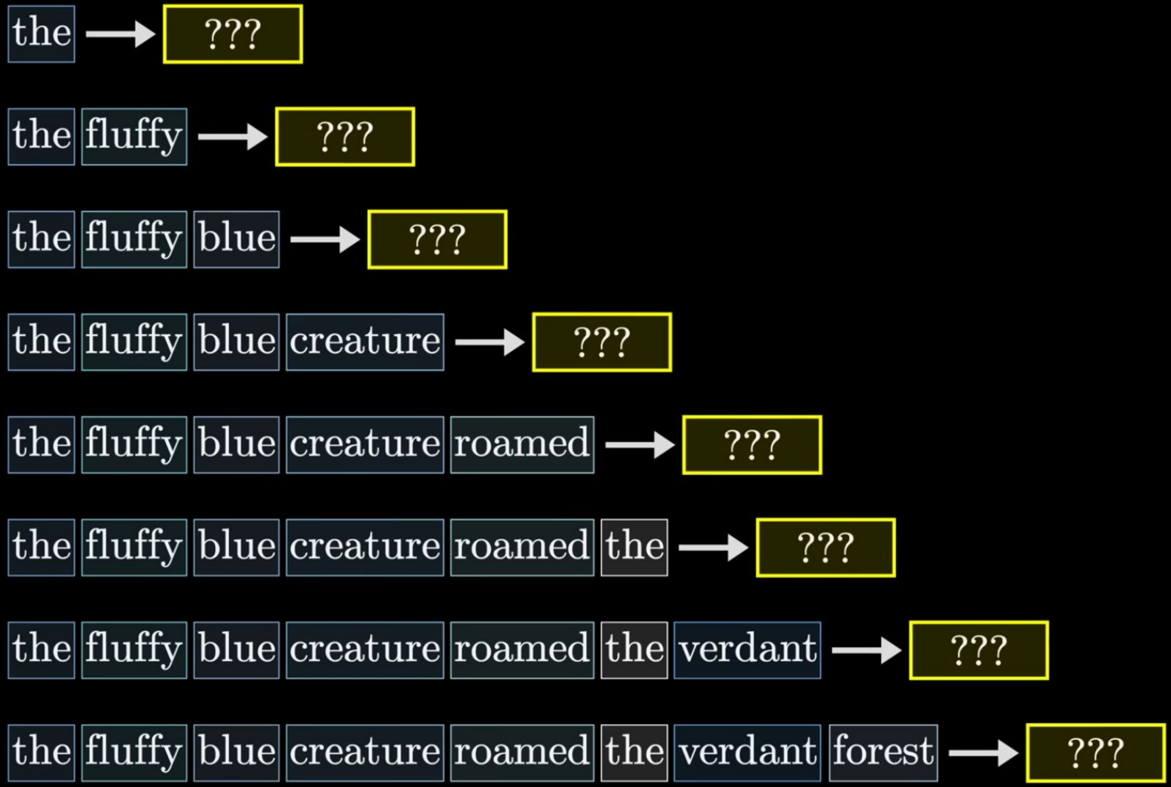
Unembedding



Objetivo: Completar a próxima palavra

To date, the cleverest thinker of all time was $\boxed{}$

Objetivo: Completar a próxima palavra

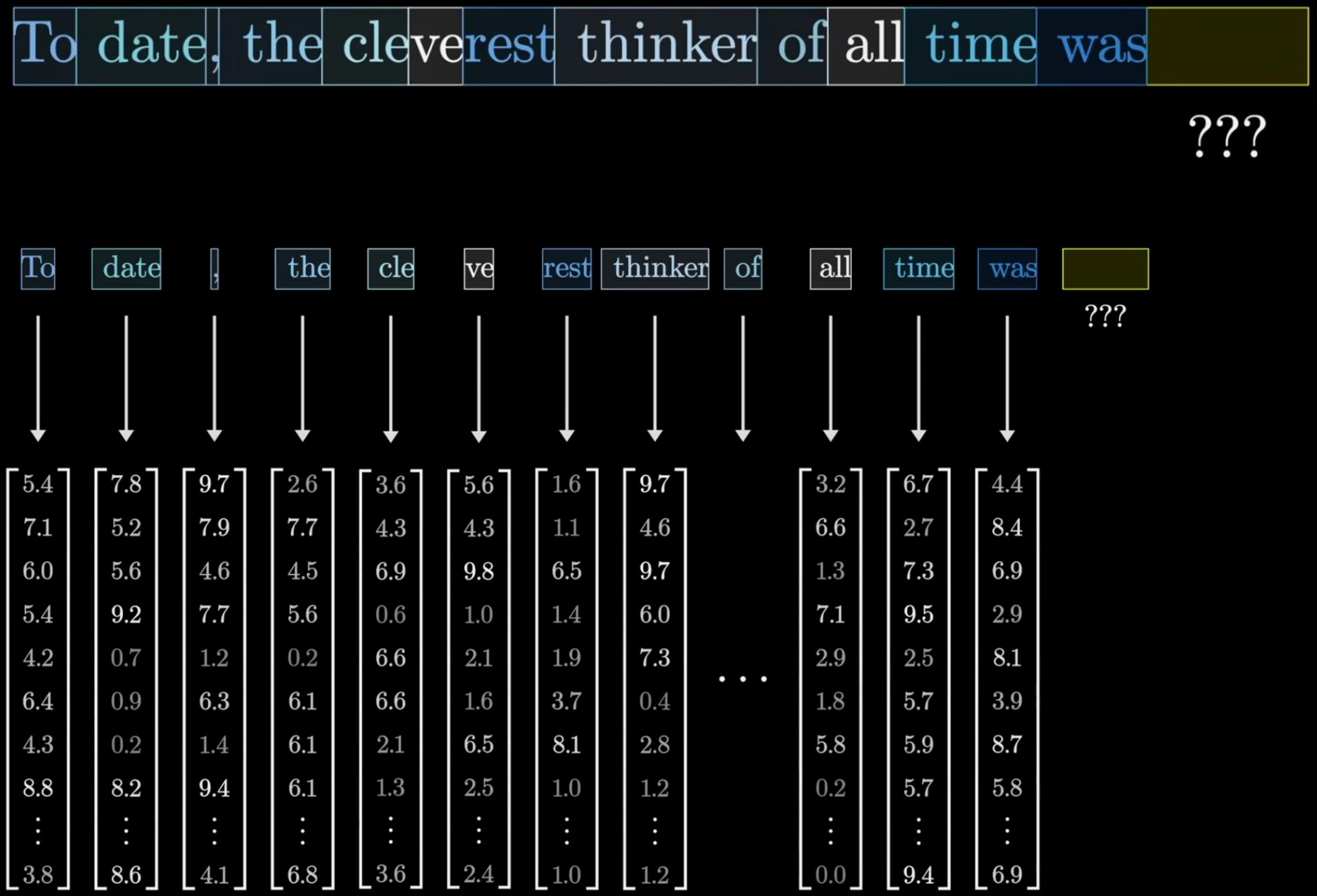


Tokens

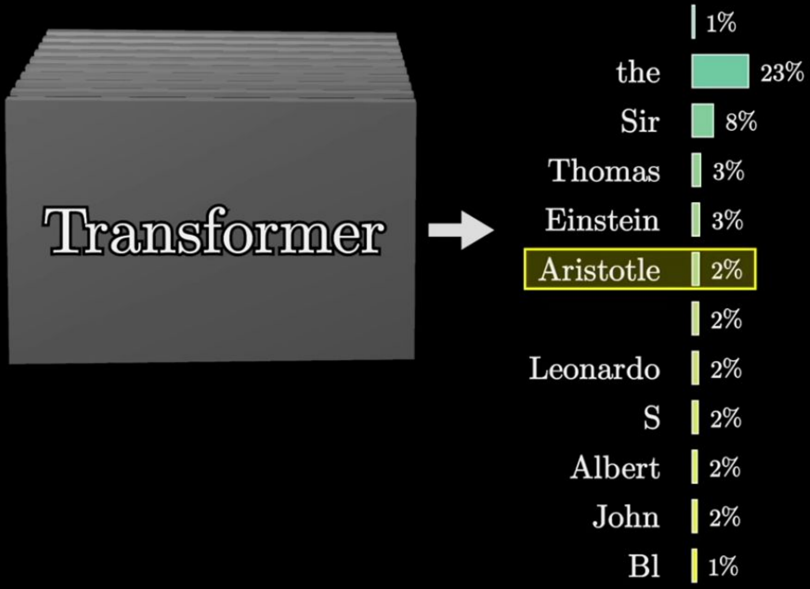
To date, the cleverest thinker of all time was

To| date|,| the| cle|ve|rest| thinker| of| all| time| was

???


To date, the cleverest thinker of all time was
???
To
date
,
the
cle
ve
rest
thinker
of
all
time
was
???
5.4 7.1 6.0 5.4 4.2 6.4 4.3 8.8 ⋮ 3.8
7.8 5.2 5.6 9.2 0.7 0.9 0.2 8.2 ⋮ 8.6
9.7 7.9 4.6 7.7 1.2 6.3 1.4 9.4 ⋮ 4.1
2.6 7.7 4.5 5.6 0.2 6.1 6.1 6.1 ⋮ 6.8
3.6 4.3 6.9 0.6 6.6 6.6 2.1 1.3 ⋮ 3.6
5.6 4.3 9.8 1.0 2.1 1.6 6.5 2.5 ⋮ 2.4
1.6 1.1 6.5 1.4 1.9 3.7 8.1 1.0 ⋮ 1.0
9.7 4.6 9.7 6.0 7.3 0.4 2.8 1.2 ⋮ 1.2
. . .
3.2 6.6 1.3 7.1 2.9 1.8 5.8 0.2 ⋮ 0.0
6.7 2.7 7.3 9.5 2.5 5.7 5.9 5.7 ⋮ 9.4
4.4 8.4 6.9 2.9 8.1 3.9 8.7 5.8 ⋮ 6.9

To date, the cleverest thinker of all time was undoubtedly ________



$$
W_E = \begin{array}{c}
\overbrace{\begin{array}{ccccccccccccccccccccc}
\text{aah} & \text{aardvark} & \text{aardwolf} & \text{aargh} & \text{ab} & \text{aback} & \text{abacterial} & \text{abacus} & \text{abalone} & \text{abandon} & \cdots & \text{zygoid} & \text{zygomatic} & \text{zygomorphic} & \text{zygosis} & \text{zygote} & \text{zygotic} & \text{zyme} & \text{zymogen} & \text{zymosis} & \text{zzz}
\end{array}}^{\text{All words, } \sim 50\text{k}} \\
\begin{bmatrix}
+1.0 & +4.3 & +2.0 & +0.9 & -1.5 & +2.9 & -1.2 & +7.8 & +9.2 & -2.3 & \cdots & +0.6 & +1.3 & +8.4 & -8.5 & -8.2 & -9.5 & +6.6 & +5.5 & +7.3 & +9.5 \\
+5.9 & -0.8 & +5.6 & -7.6 & +2.8 & -7.1 & +8.8 & +0.4 & -1.7 & -4.7 & \cdots & -0.9 & +1.4 & -9.5 & +2.3 & +2.2 & +2.3 & +8.8 & +3.6 & -2.8 & -1.2 \\
+3.9 & -8.7 & +3.3 & +3.4 & -5.7 & -7.3 & -3.7 & -2.7 & +1.4 & -1.2 & \cdots & -7.9 & -5.8 & -6.7 & +3.0 & -4.9 & -0.7 & -5.1 & -6.8 & -7.7 & +3.1 \\
-7.2 & -6.0 & -2.6 & +6.4 & -8.0 & +6.7 & -8.0 & +9.4 & -0.6 & +9.4 & \cdots & +4.7 & -9.1 & -4.3 & -7.5 & -4.0 & -7.5 & -3.6 & -1.7 & -8.6 & +3.8 \\
+1.3 & -4.6 & +0.5 & -8.0 & +1.5 & +8.5 & -3.6 & +3.3 & -7.3 & +4.3 & \cdots & -6.3 & +1.7 & -9.5 & +6.5 & -9.8 & +3.5 & -4.6 & +4.7 & +9.2 & -5.0 \\
+1.5 & +1.8 & +1.4 & -5.5 & +9.0 & -1.0 & +6.9 & +3.9 & -4.0 & +6.2 & \cdots & +7.5 & +1.6 & +7.6 & +3.8 & +4.5 & +0.0 & +9.0 & +2.9 & -1.5 & +2.1 \\
-9.5 & -3.9 & +3.2 & -4.2 & +2.3 & -1.4 & -7.2 & -4.0 & +1.4 & +1.8 & \cdots & +3.0 & +3.0 & -1.4 & +7.9 & -2.6 & -1.3 & +7.8 & +6.1 & +4.0 & -7.9 \\
+8.3 & +4.2 & +9.9 & -6.9 & +7.3 & -6.7 & +2.3 & -7.4 & +6.9 & +6.1 & \cdots & -1.8 & -8.5 & +3.9 & -0.9 & +4.4 & +7.3 & +9.4 & +7.0 & -9.7 & -2.8 \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
-3.7 & -2.0 & -5.7 & -6.2 & +8.8 & +4.7 & -0.2 & -5.4 & -4.9 & -8.8 & \cdots & -3.7 & +3.9 & -2.4 & -6.3 & -9.4 & -8.6 & +3.6 & -0.9 & +0.7 & +7.9
\end{bmatrix}
\end{array}
$$

Embedding matrix

All words, $\sim 50\text{k}$

$W_E =$

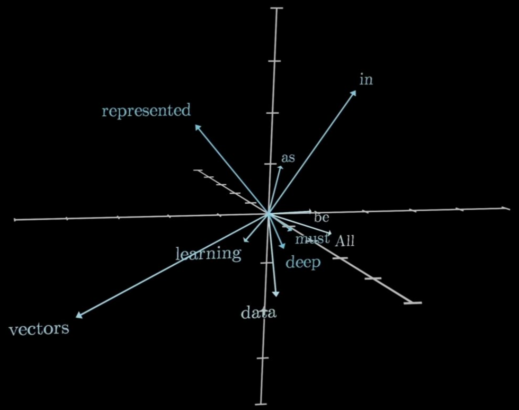
aah	aardvark	aardwolf	aargh	ab	aback	abacterial	abacus	abalone	abandon	…	zygoid	zygomatic	zygomorphic	zygosis	zygote	zygotic	zyme	zymogen	zymosis	zzz
+1.0	+4.3	+2.0	+0.9	−1.5	+2.9	−1.2	+7.8	+9.2	−2.3	⋯	+0.6	+1.3	+8.4	−8.5	−8.2	−9.5	+6.6	+5.5	+7.3	+9.5
+5.9	−0.8	+5.6	−7.6	+2.8	−7.1	+8.8	+0.4	−1.7	−4.7	⋯	−0.9	+1.4	−9.5	+2.3	+2.2	+2.3	+8.8	+3.6	−2.8	−1.2
+3.9	−8.7	+3.3	+3.4	−5.7	−7.3	−3.7	−2.7	+1.4	−1.2	⋯	−7.9	−5.8	−6.7	+3.0	−4.9	−0.7	−5.1	−6.8	−7.7	+3.1
−7.2	−6.0	−2.6	+6.4	−8.0	+6.7	−8.0	+9.4	−0.6	+9.4	⋯	+4.7	−9.1	−4.3	−7.5	−4.0	−7.5	−3.6	−1.7	−8.6	+3.8
+1.3	−4.6	+0.5	−8.0	+1.5	+8.5	−3.6	+3.3	−7.3	+4.3	⋯	−6.3	+1.7	−9.5	+6.5	−9.8	+3.5	−4.6	+4.7	+9.2	−5.0
+1.5	+1.8	+1.4	−5.5	+9.0	−1.0	+6.9	+3.9	−4.0	+6.2	⋯	+7.5	+1.6	+7.6	+3.8	+4.5	+0.0	+9.0	+2.9	−1.5	+2.1
−9.5	−3.9	+3.2	−4.2	+2.3	−1.4	−7.2	−4.0	+1.4	+1.8	⋯	+3.0	+3.0	−1.4	+7.9	−2.6	−1.3	+7.8	+6.1	+4.0	−7.9
+8.3	+4.2	+9.9	−6.9	+7.3	−6.7	+2.3	−7.4	+6.9	+6.1	⋯	−1.8	−8.5	+3.9	−0.9	+4.4	+7.3	+9.4	+7.0	−9.7	−2.8
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋱	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
−3.7	−2.0	−5.7	−6.2	+8.8	+4.7	−0.2	−5.4	−4.9	−8.8	⋯	−3.7	+3.9	−2.4	−6.3	−9.4	−8.6	+3.6	−0.9	+0.7	+7.9

Embedding matrix

Word Vectors - 12.288 (GPT3)

Words $\xrightarrow{\text{“Embedding”}}$ Vectors

All
data
in
deep
learning
must
be
represented
as
vectors



https://youtu.be/wjZofJX0v4M?si=QfjMfnixv-V--ymA&t=914

50,257 tokens

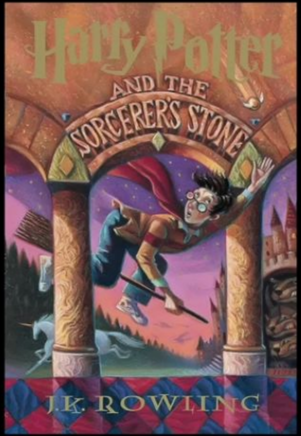
12,288

<table>
<tr><th>!</th><th>"</th><th>#</th><th>$</th><th>%</th><th>&</th><th>'</th><th>(</th><th>)</th><th>*</th><th>⋯</th><th>\</th><th>."</th><th>Compar</th><th>amplification</th><th>ominated</th><th>regress</th><th>Collider</th><th>informants</th><th>gazed</th><th><|endoftext|></th></tr>
<tr><td>−4.0</td><td>+1.0</td><td>+8.5</td><td>+0.4</td><td>−4.6</td><td>+7.5</td><td>−2.5</td><td>−9.9</td><td>−5.0</td><td>−3.6</td><td>⋯</td><td>+7.1</td><td>−0.8</td><td>−1.1</td><td>−3.2</td><td>+7.5</td><td>+8.8</td><td>+9.7</td><td>−2.4</td><td>+9.2</td><td>+5.8</td></tr>
<tr><td>+3.5</td><td>−5.1</td><td>−5.6</td><td>−6.6</td><td>+8.4</td><td>−4.1</td><td>−0.9</td><td>−0.1</td><td>+5.5</td><td>+6.8</td><td>⋯</td><td>−7.1</td><td>−1.4</td><td>+6.8</td><td>+6.3</td><td>−7.9</td><td>−6.8</td><td>−3.9</td><td>−8.4</td><td>−1.5</td><td>−7.8</td></tr>
<tr><td>+1.4</td><td>−5.0</td><td>+1.9</td><td>−7.6</td><td>+9.4</td><td>+8.6</td><td>−2.1</td><td>−5.1</td><td>−4.9</td><td>−0.3</td><td>⋯</td><td>−9.1</td><td>+2.8</td><td>−1.8</td><td>−2.4</td><td>+6.1</td><td>+4.1</td><td>+9.0</td><td>−2.9</td><td>+7.9</td><td>+5.3</td></tr>
<tr><td>−2.8</td><td>+2.4</td><td>−4.2</td><td>+7.4</td><td>−7.7</td><td>−5.7</td><td>−6.3</td><td>−1.9</td><td>+4.9</td><td>+0.5</td><td>⋯</td><td>−0.2</td><td>−9.9</td><td>−1.5</td><td>−8.6</td><td>−5.8</td><td>+8.6</td><td>−5.6</td><td>+7.1</td><td>+6.0</td><td>−6.7</td></tr>
<tr><td>+2.1</td><td>−7.6</td><td>+4.5</td><td>+2.7</td><td>+6.2</td><td>−0.4</td><td>+8.2</td><td>−8.9</td><td>−4.1</td><td>+4.3</td><td>⋯</td><td>−1.6</td><td>−6.5</td><td>−7.8</td><td>+6.3</td><td>−0.5</td><td>+7.6</td><td>+4.6</td><td>−1.8</td><td>−2.5</td><td>+0.3</td></tr>
<tr><td>+7.7</td><td>+4.7</td><td>−9.8</td><td>+3.8</td><td>+8.3</td><td>+4.2</td><td>−6.4</td><td>−0.3</td><td>−7.1</td><td>−2.8</td><td>⋯</td><td>+8.7</td><td>+8.4</td><td>−4.3</td><td>−3.2</td><td>+2.0</td><td>+9.2</td><td>−7.0</td><td>−4.8</td><td>+7.4</td><td>−0.2</td></tr>
<tr><td>+7.9</td><td>−6.2</td><td>+0.6</td><td>−3.4</td><td>−3.6</td><td>−1.1</td><td>−1.3</td><td>−2.8</td><td>+8.2</td><td>+4.6</td><td>⋯</td><td>+4.5</td><td>−4.2</td><td>+1.5</td><td>+5.5</td><td>+5.9</td><td>−3.1</td><td>+5.4</td><td>+4.7</td><td>−7.1</td><td>+7.2</td></tr>
<tr><td>−1.2</td><td>−0.3</td><td>−1.0</td><td>+1.3</td><td>+2.4</td><td>+0.0</td><td>+7.3</td><td>+2.5</td><td>−2.0</td><td>−1.6</td><td>⋯</td><td>+6.2</td><td>−3.0</td><td>−5.7</td><td>−8.7</td><td>+7.4</td><td>+8.3</td><td>−7.5</td><td>−3.3</td><td>−6.4</td><td>−7.6</td></tr>
<tr><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋱</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td><td>⋮</td></tr>
<tr><td>+7.9</td><td>−8.8</td><td>+9.5</td><td>−8.0</td><td>+7.2</td><td>+1.3</td><td>−2.6</td><td>−3.1</td><td>+5.1</td><td>−3.7</td><td>⋯</td><td>+3.1</td><td>+0.3</td><td>−0.3</td><td>+7.9</td><td>+1.1</td><td>+6.5</td><td>+4.5</td><td>−9.1</td><td>+5.4</td><td>−5.6</td></tr>
</table>

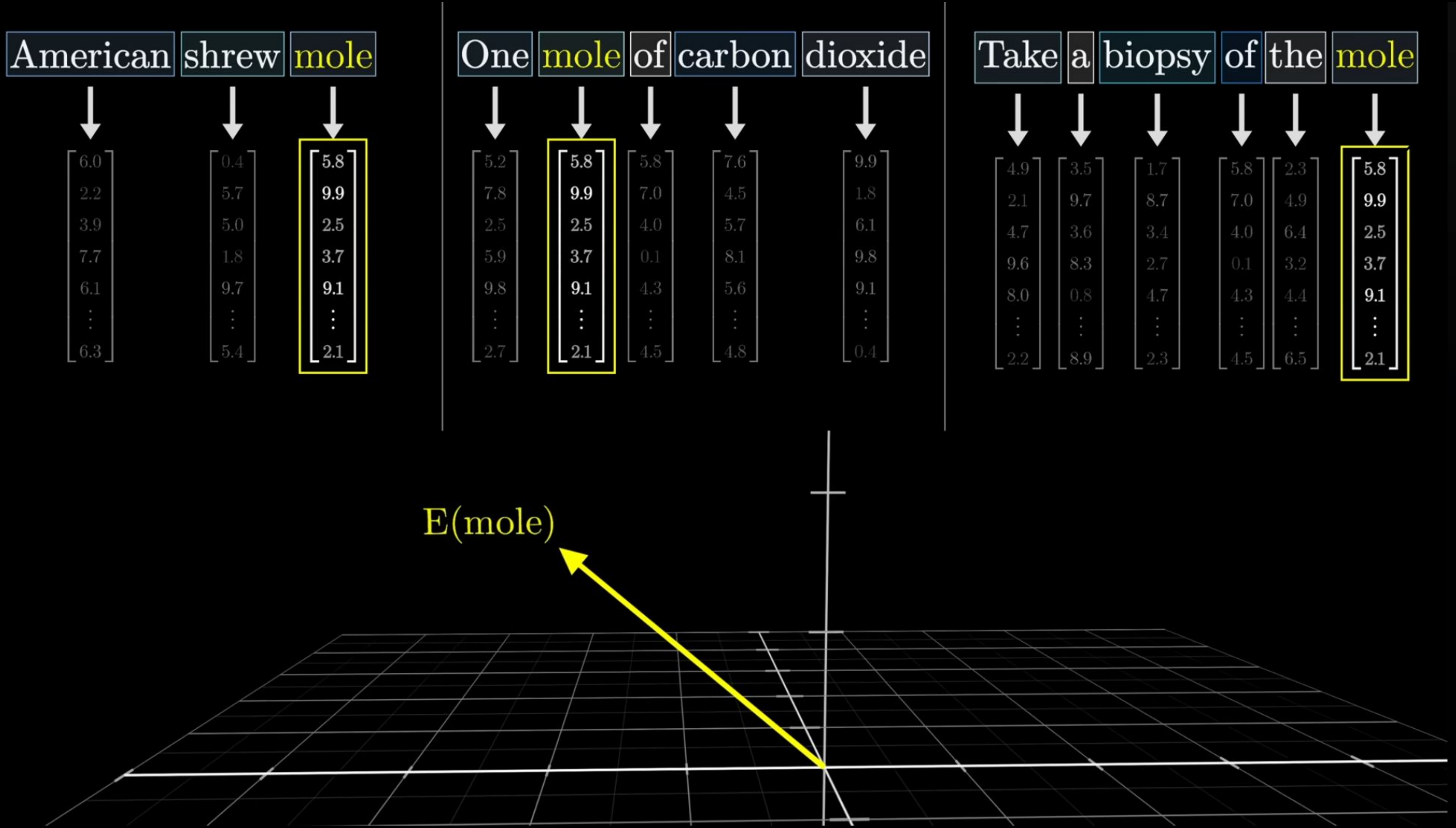
$$W_E = \text{Embedding matrix}$$

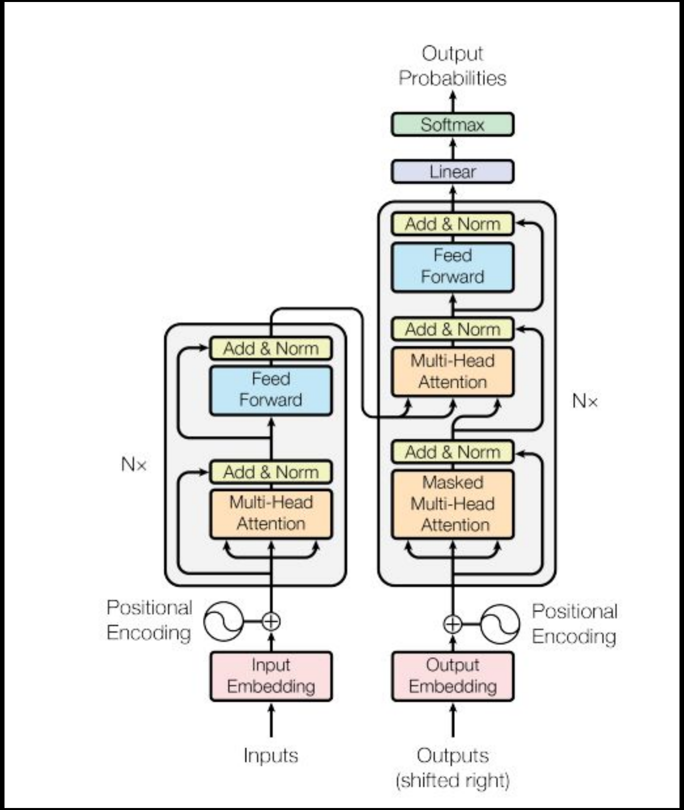
Embeddings = Significado das "palavras"

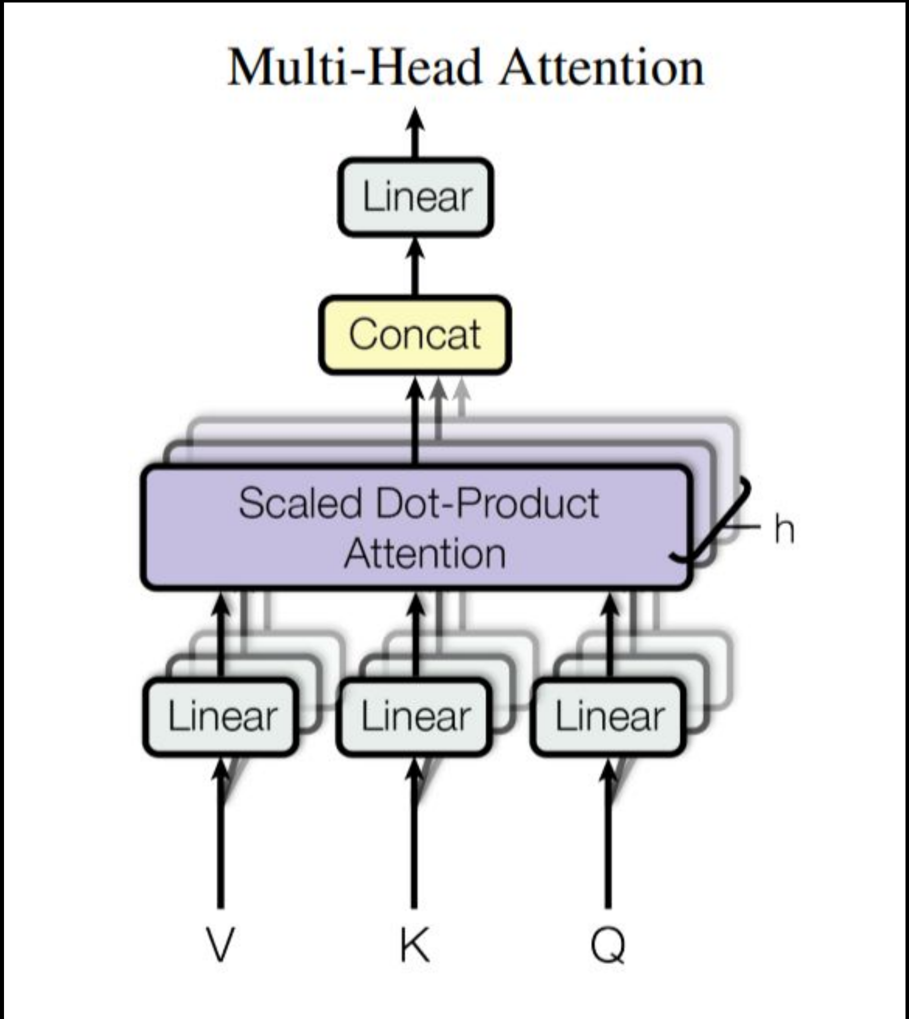
... wizard ... Hogwarts ... Hermione ... Harry

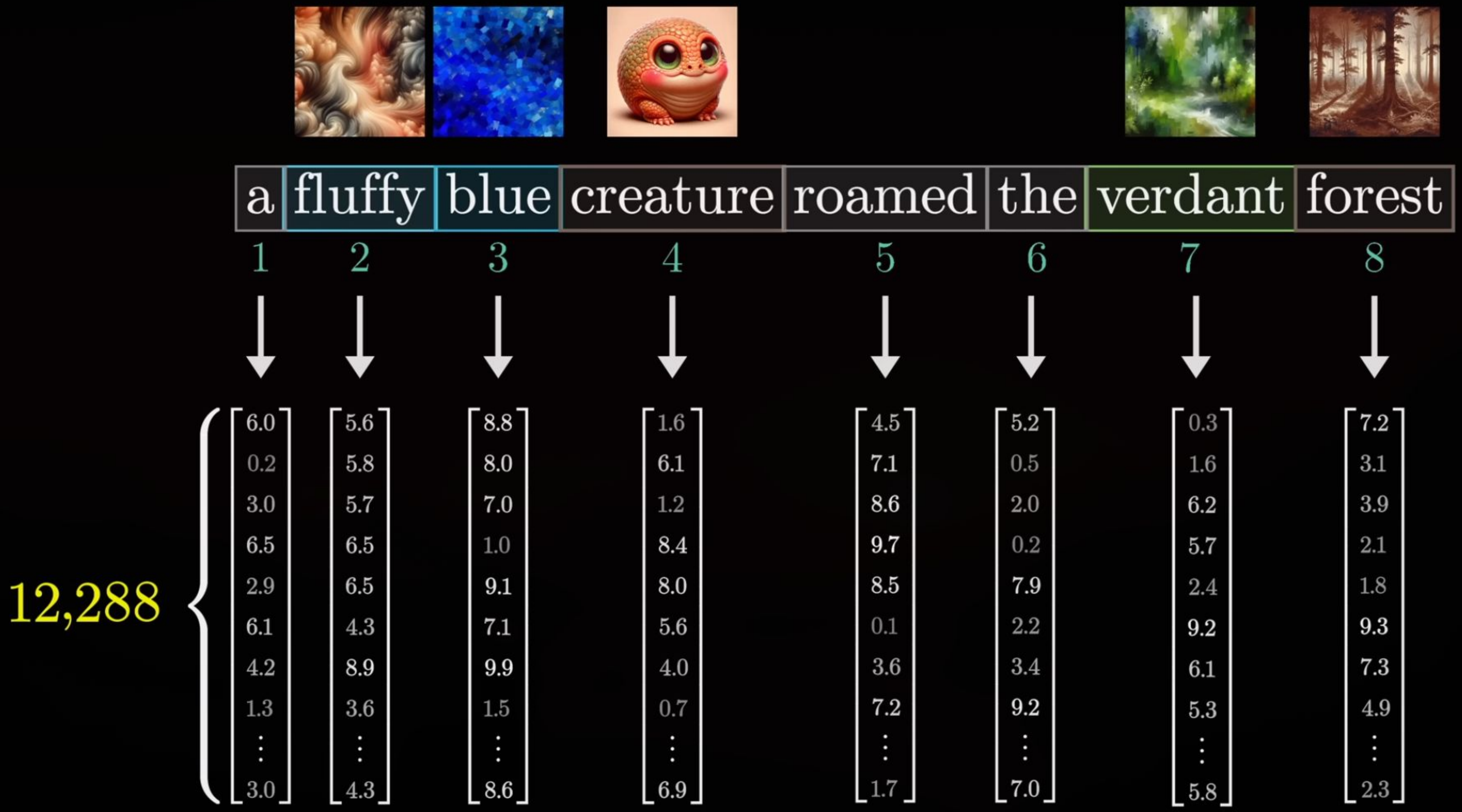


... Queen ... Sussex ... William ... Harry

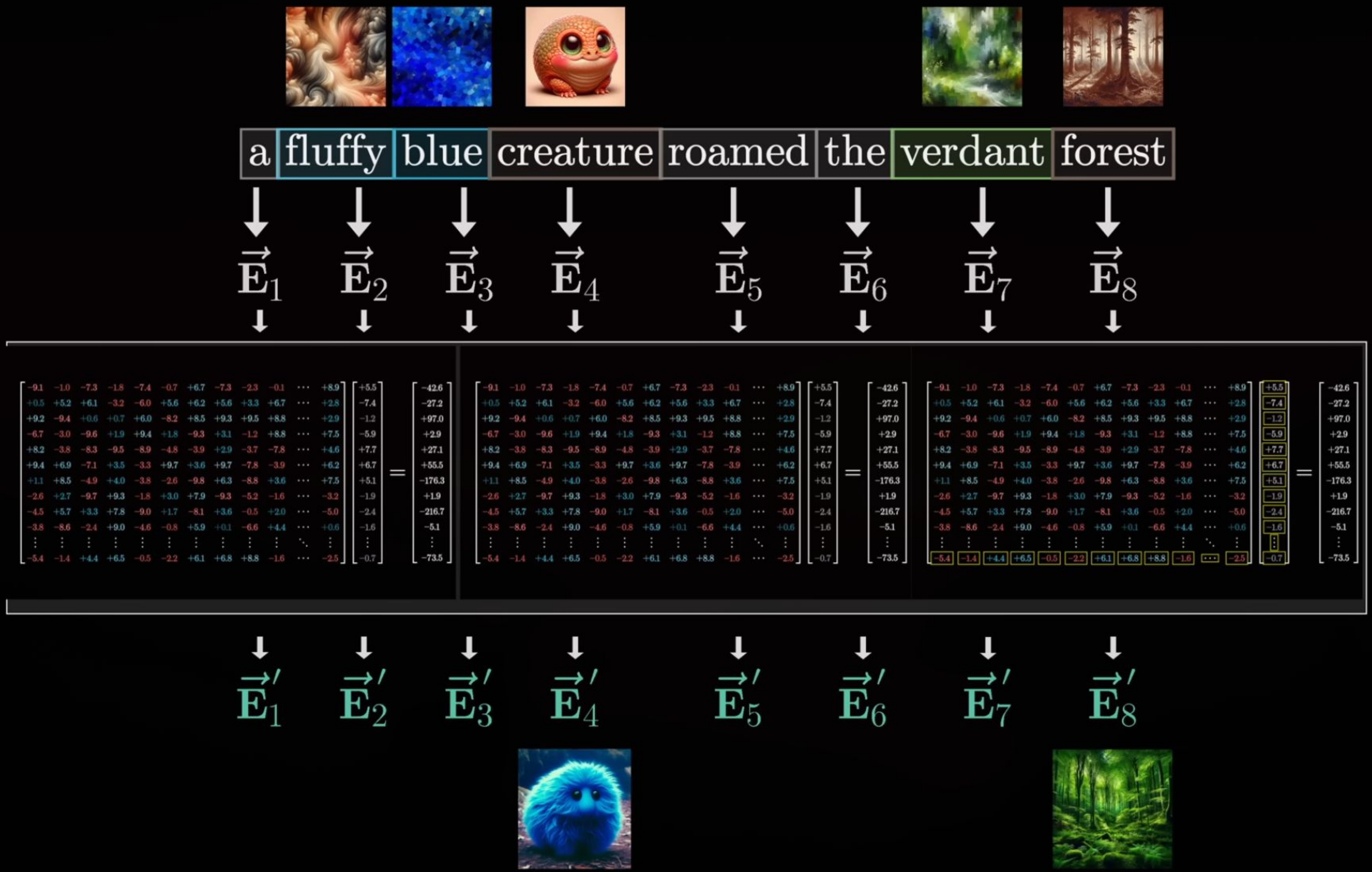

American shrew mole
6.0 2.2 3.9 7.7 6.1 ⋮ 6.3
0.4 5.7 5.0 1.8 9.7 ⋮ 5.4
5.8 9.9 2.5 3.7 9.1 ⋮ 2.1
One mole of carbon dioxide
5.2 7.8 2.5 5.9 9.8 ⋮ 2.7
5.8 9.9 2.5 3.7 9.1 ⋮ 2.1
5.8 7.0 4.0 0.1 4.3 ⋮ 4.5
7.6 4.5 5.7 8.1 5.6 ⋮ 4.8
9.9 1.8 6.1 9.8 9.1 ⋮ 0.4
Take a biopsy of the mole
4.9 2.1 4.7 9.6 8.0 ⋮ 2.2
3.5 9.7 3.6 8.3 0.8 ⋮ 8.9
1.7 8.7 3.4 2.7 4.7 ⋮ 2.3
5.8 7.0 4.0 0.1 4.3 ⋮ 4.5
2.3 4.9 6.4 3.2 4.4 ⋮ 6.5
5.8 9.9 2.5 3.7 9.1 ⋮ 2.1
E(mole)


Output
Probabilities
Softmax
Linear
Add & Norm
Feed
Forward
Add & Norm
Multi-Head
Attention
Add & Norm
Masked
Multi-Head
Attention
Nx
Add & Norm
Feed
Forward
Add & Norm
Multi-Head
Attention
Nx
Positional
Encoding
Input
Embedding
Output
Embedding
Positional
Encoding
Inputs
Outputs
(shifted right)


Multi-Head Attention
Linear
Concat
Scaled Dot-Product
Attention
h
Linear
Linear
Linear
V
K
Q



a
fluffy
blue
creature
roamed
the
verdant
forest
1
2
3
4
5
6
7
8
12,288
6.0 0.2 3.0 6.5 2.9 6.1 4.2 1.3 ⋮ 3.0
5.6 5.8 5.7 6.5 6.5 4.3 8.9 3.6 ⋮ 4.3
8.8 8.0 7.0 1.0 9.1 7.1 9.9 1.5 ⋮ 8.6
1.6 6.1 1.2 8.4 8.0 5.6 4.0 0.7 ⋮ 6.9
4.5 7.1 8.6 9.7 8.5 0.1 3.6 7.2 ⋮ 1.7
5.2 0.5 2.0 0.2 7.9 2.2 3.4 9.2 ⋮ 7.0
0.3 1.6 6.2 5.7 2.4 9.2 6.1 5.3 ⋮ 5.8
7.2 3.1 3.9 2.1 1.8 9.3 7.3 4.9 ⋮ 2.3



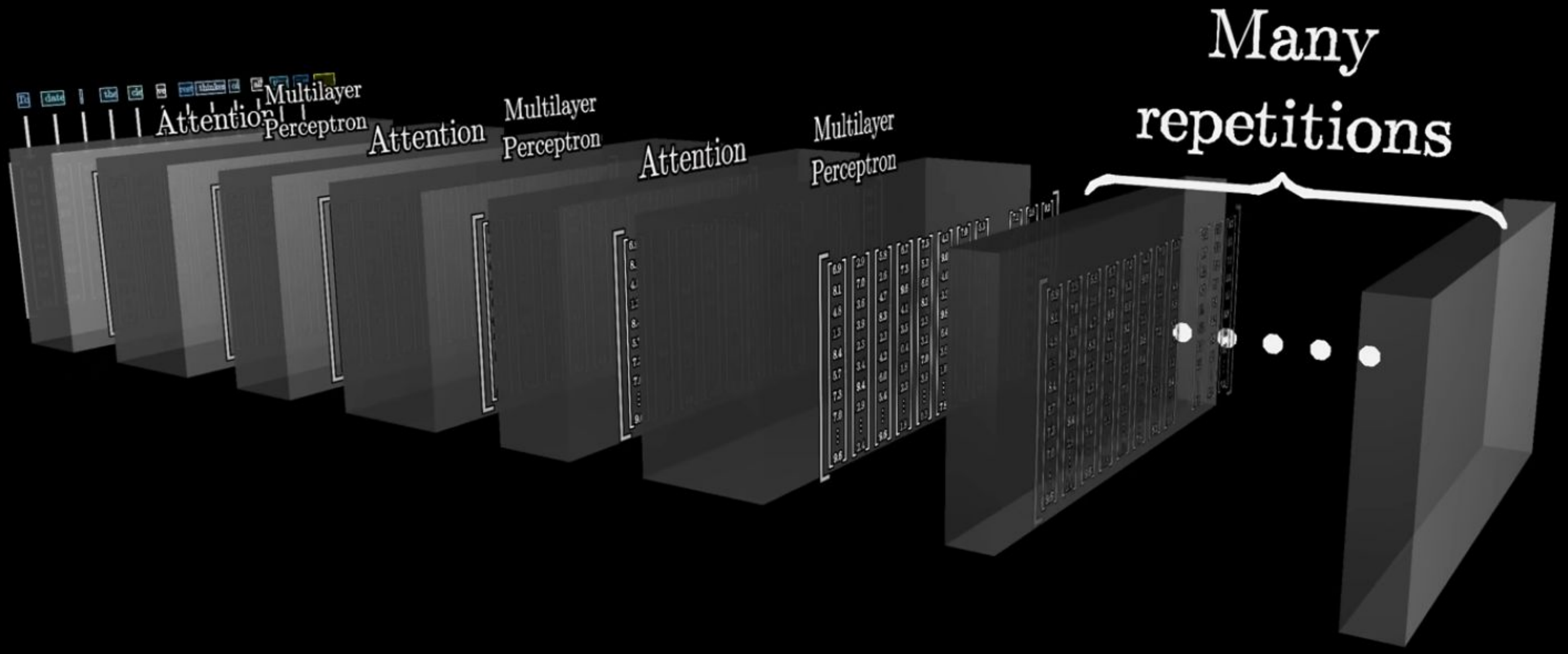
a
fluffy
blue
creature
roamed
the
verdant
forest
$\vec{E}_1$ $\vec{E}_2$ $\vec{E}_3$ $\vec{E}_4$ $\vec{E}_5$ $\vec{E}_6$ $\vec{E}_7$ $\vec{E}_8$
$\vec{E}_1'$ $\vec{E}_2'$ $\vec{E}_3'$ $\vec{E}_4'$ $\vec{E}_5'$ $\vec{E}_6'$ $\vec{E}_7'$ $\vec{E}_8'$

	a $\downarrow$ $\vec{E}_1 \xrightarrow{W_Q} \vec{Q}_1$	fluffy $\downarrow$ $\vec{E}_2 \xrightarrow{W_Q} \vec{Q}_2$	blue $\downarrow$ $\vec{E}_3 \xrightarrow{W_Q} \vec{Q}_3$	creature $\downarrow$ $\vec{E}_4 \xrightarrow{W_Q} \vec{Q}_4$	roamed $\downarrow$ $\vec{E}_5 \xrightarrow{W_Q} \vec{Q}_5$	the $\downarrow$ $\vec{E}_6 \xrightarrow{W_Q} \vec{Q}_6$	verdant $\downarrow$ $\vec{E}_7 \xrightarrow{W_Q} \vec{Q}_7$	forest $\downarrow$ $\vec{E}_8 \xrightarrow{W_Q} \vec{Q}_8$
a $\rightarrow \vec{E}_1 \xrightarrow{W_k} \vec{K}_1$	$\vec{K}_1 \cdot \vec{Q}_1$	$\vec{K}_1 \cdot \vec{Q}_2$	$\vec{K}_1 \cdot \vec{Q}_3$	$\vec{K}_1 \cdot \vec{Q}_4$	$\vec{K}_1 \cdot \vec{Q}_5$	$\vec{K}_1 \cdot \vec{Q}_6$	$\vec{K}_1 \cdot \vec{Q}_7$	$\vec{K}_1 \cdot \vec{Q}_8$
fluffy $\rightarrow \vec{E}_2 \xrightarrow{W_k} \vec{K}_2$	$\vec{K}_2 \cdot \vec{Q}_1$	$\vec{K}_2 \cdot \vec{Q}_2$	$\vec{K}_2 \cdot \vec{Q}_3$	$\vec{K}_2 \cdot \vec{Q}_4$	$\vec{K}_2 \cdot \vec{Q}_5$	$\vec{K}_2 \cdot \vec{Q}_6$	$\vec{K}_2 \cdot \vec{Q}_7$	$\vec{K}_2 \cdot \vec{Q}_8$
blue $\rightarrow \vec{E}_3 \xrightarrow{W_k} \vec{K}_3$	$\vec{K}_3 \cdot \vec{Q}_1$	$\vec{K}_3 \cdot \vec{Q}_2$	$\vec{K}_3 \cdot \vec{Q}_3$	$\vec{K}_3 \cdot \vec{Q}_4$	$\vec{K}_3 \cdot \vec{Q}_5$	$\vec{K}_3 \cdot \vec{Q}_6$	$\vec{K}_3 \cdot \vec{Q}_7$	$\vec{K}_3 \cdot \vec{Q}_8$
creature $\rightarrow \vec{E}_4 \xrightarrow{W_k} \vec{K}_4$	$\vec{K}_4 \cdot \vec{Q}_1$	$\vec{K}_4 \cdot \vec{Q}_2$	$\vec{K}_4 \cdot \vec{Q}_3$	$\vec{K}_4 \cdot \vec{Q}_4$	$\vec{K}_4 \cdot \vec{Q}_5$	$\vec{K}_4 \cdot \vec{Q}_6$	$\vec{K}_4 \cdot \vec{Q}_7$	$\vec{K}_4 \cdot \vec{Q}_8$
roamed $\rightarrow \vec{E}_5 \xrightarrow{W_k} \vec{K}_5$	$\vec{K}_5 \cdot \vec{Q}_1$	$\vec{K}_5 \cdot \vec{Q}_2$	$\vec{K}_5 \cdot \vec{Q}_3$	$\vec{K}_5 \cdot \vec{Q}_4$	$\vec{K}_5 \cdot \vec{Q}_5$	$\vec{K}_5 \cdot \vec{Q}_6$	$\vec{K}_5 \cdot \vec{Q}_7$	$\vec{K}_5 \cdot \vec{Q}_8$
the $\rightarrow \vec{E}_6 \xrightarrow{W_k} \vec{K}_6$	$\vec{K}_6 \cdot \vec{Q}_1$	$\vec{K}_6 \cdot \vec{Q}_2$	$\vec{K}_6 \cdot \vec{Q}_3$	$\vec{K}_6 \cdot \vec{Q}_4$	$\vec{K}_6 \cdot \vec{Q}_5$	$\vec{K}_6 \cdot \vec{Q}_6$	$\vec{K}_6 \cdot \vec{Q}_7$	$\vec{K}_6 \cdot \vec{Q}_8$
verdant $\rightarrow \vec{E}_7 \xrightarrow{W_k} \vec{K}_7$	$\vec{K}_7 \cdot \vec{Q}_1$	$\vec{K}_7 \cdot \vec{Q}_2$	$\vec{K}_7 \cdot \vec{Q}_3$	$\vec{K}_7 \cdot \vec{Q}_4$	$\vec{K}_7 \cdot \vec{Q}_5$	$\vec{K}_7 \cdot \vec{Q}_6$	$\vec{K}_7 \cdot \vec{Q}_7$	$\vec{K}_7 \cdot \vec{Q}_8$
forest $\rightarrow \vec{E}_8 \xrightarrow{W_k} \vec{K}_8$	$\vec{K}_8 \cdot \vec{Q}_1$	$\vec{K}_8 \cdot \vec{Q}_2$	$\vec{K}_8 \cdot \vec{Q}_3$	$\vec{K}_8 \cdot \vec{Q}_4$	$\vec{K}_8 \cdot \vec{Q}_5$	$\vec{K}_8 \cdot \vec{Q}_6$	$\vec{K}_8 \cdot \vec{Q}_7$	$\vec{K}_8 \cdot \vec{Q}_8$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{\boxed{K^T Q}}{\sqrt{d_k}}\right) V$$

	Q_1	Q_2	Q_3	Q_4	Q_5	$\cdots$	Q_n
K_1	$Q_1 \cdot K_1$	$Q_2 \cdot K_1$	$Q_3 \cdot K_1$	$Q_4 \cdot K_1$	$Q_5 \cdot K_1$	$\cdots$	$Q_n \cdot K_1$
K_2	$Q_1 \cdot K_2$	$Q_2 \cdot K_2$	$Q_3 \cdot K_2$	$Q_4 \cdot K_2$	$Q_5 \cdot K_2$	$\cdots$	$Q_n \cdot K_2$
K_3	$Q_1 \cdot K_3$	$Q_2 \cdot K_3$	$Q_3 \cdot K_3$	$Q_4 \cdot K_3$	$Q_5 \cdot K_3$	$\cdots$	$Q_n \cdot K_3$
K_4	$Q_1 \cdot K_4$	$Q_2 \cdot K_4$	$Q_3 \cdot K_4$	$Q_4 \cdot K_4$	$Q_5 \cdot K_4$	$\cdots$	$Q_n \cdot K_4$
K_5	$Q_1 \cdot K_5$	$Q_2 \cdot K_5$	$Q_3 \cdot K_5$	$Q_4 \cdot K_5$	$Q_5 \cdot K_5$	$\cdots$	$Q_n \cdot K_5$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$

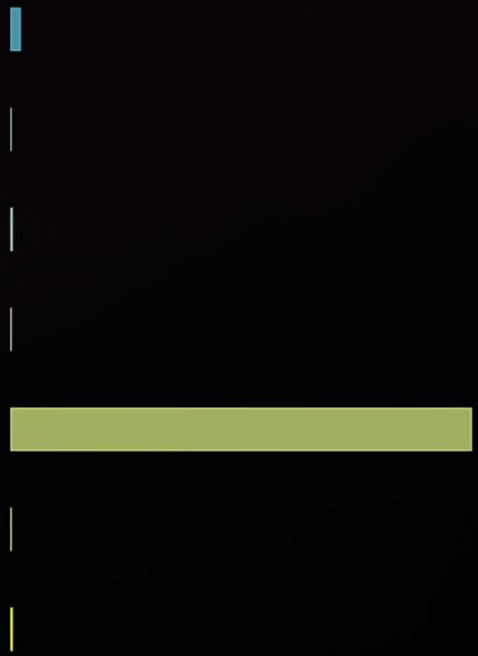
Attention Pattern


Attention
Multilayer Perceptron
Attention
Multilayer Perceptron
Attention
Multilayer Perceptron
Many repetitions

softmax with temperature

$T = 1.00$

$$\begin{bmatrix} +6.0 \\ -5.0 \\ +4.0 \\ +1.5 \\ +9.9 \\ -2.3 \\ +4.0 \end{bmatrix} \longrightarrow \begin{bmatrix} e^{x_1/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \\ e^{x_2/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \\ e^{x_3/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \\ e^{x_4/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \\ e^{x_5/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \\ e^{x_6/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \\ e^{x_7/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \end{bmatrix} = \begin{bmatrix} 0.02 \\ 0.00 \\ 0.00 \\ 0.00 \\ 0.97 \\ 0.00 \\ 0.00 \end{bmatrix}$$



softmax with temperature

$T = 9.18$

$$\begin{bmatrix} +6.0 \\ -5.0 \\ +4.0 \\ +1.5 \\ +9.9 \\ -2.3 \\ +4.0 \end{bmatrix} \longrightarrow \begin{bmatrix} e^{x_1/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \\ e^{x_2/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \\ e^{x_3/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \\ e^{x_4/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \\ e^{x_5/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \\ e^{x_6/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \\ e^{x_7/T} \Big/ \sum_{n=0}^{N-1} e^{x_n/T} \end{bmatrix} = \begin{bmatrix} 0.18 \\ 0.06 \\ 0.15 \\ 0.11 \\ 0.28 \\ 0.07 \\ 0.15 \end{bmatrix}$$